

Ass 3

Credit Points $10 \times 4 = 40$

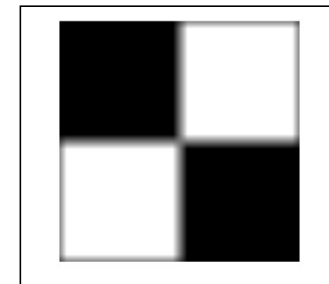
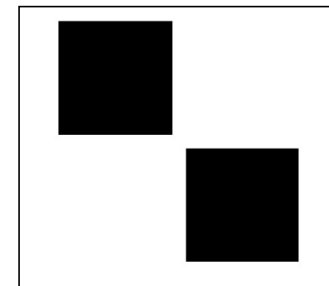
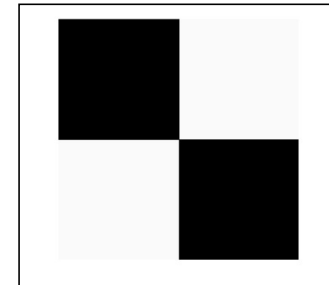
CLO 1, 2 credits = [20, 20]

Spatial Filtering (Neighborhood Processing)

Q1: Linear Spatial Filtering:

```
>> f = imread('boxes.tif');  
>> class(f)  
ans =  
      uint8  
>> w = ones(31);  
>> g = imfilter(f, w, 'replicate');  
>> imshow(f), figure, imshow(g, [ ])  
  
>> fd = im2double(f);  
>> gd = imfilter(fd, w);  
>> imshow(fd), figure, imshow(gd, [ ])
```

Why?



Spatial Filtering (Neighborhood Processing)

Linear Spatial Filtering

Q2: a: Explain the following

- i. Why image boxes.tif is not blurred around the outside boundary of black boxes?
- ii. Why corners of black boxes don't join when we use uint8 data type of image?

b: Consider an image $f = [1 \ 0 \ 0 \ 0 \ 0; 0 \ 2 \ 0 \ 0 \ 0; 0 \ 0 \ 3 \ 0 \ 0; 0 \ 0 \ 0 \ 4 \ 0; 0 \ 0 \ 0 \ 0 \ 5]$.

Filter it by hand with mask $w = [-1 \ -1 \ -1; -1 \ 8 \ -1; -1 \ -1 \ -1]$ with padding of

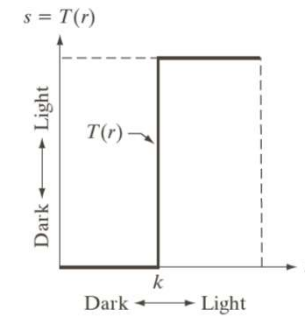
- i. Zeroes
- ii. Symmetric
- iii. Replicate
- iv. Circular

Verify your results through Matlab.

#Spatial Filtering (Neighborhood Processing)

Linear Spatial Filtering

Q3:



a) In a character recognition application, text pages are reduced to binary form using a thresholding transformation function of the form shown in figure. This is followed by a procedure that thins the characters until they become strings of binary 1's on a background of 0's. Due to noise, the binarization and thinning processes result in broken strings of characters with gaps ranging from 1 to 3 pixels. One way to “repair” the gaps is to run an averaging mask over the binary image to blur it, and thus create bridges of nonzero pixels between gaps. Give the (odd) size of the smallest averaging mask capable of performing this task.

(b) After bridging the gaps, it is desired to threshold the image in order to convert it back to binary form. For your answer in (a), what is the minimum value of the threshold required to accomplish this, without causing the segments to break up again.

#Spatial Filtering (Neighborhood Processing)

Linear Spatial Filtering

Q4:

The three images shown were blurred using square averaging masks of sizes $n = 23$, 25 , and 45 , respectively. The vertical bars on the left lower part of (a) and (c) are blurred, but a clear separation exists between them. However, the bars have merged in image (b), in spite of the fact that the mask that produced this image is significantly smaller than the mask that produced image (c). Explain the reason for this.



Hint: Vertical bars are 5 pixels wide and their separation is 20 pixels